

Sai Kumar M

DATA ANALYST

WI, USA | +1 (856) 605-7716 | saikumarmallepaka14@gmail.com | [LinkedIn](#)

SUMMARY

- Data Analyst with **4+ years of experience** supporting business teams by converting requirements into **functional documentation, process flows**, and clear reporting outputs that drive daily decisions.
- Built and maintained analytics workflows using **SQL** and **Python**, integrating insights into **Power BI** and **Tableau** for KPI tracking and stakeholder reporting.
- Hands-on experience with **data profiling, validation, migration, conversion, and integration**, improving data accuracy through structured **data modeling** and governance standards.
- Delivered interactive **Power BI dashboards** using **Power Query, Power Pivot, and Power View**, delivering drill-down reporting and performance visibility for leadership teams.
- Strong SQL optimization skills across **SQL Server, Oracle, MySQL, and PostgreSQL**, with practical exposure to **SAS, Git/GitHub, Apache Kafka, REST APIs**, and Azure services such as **ADF, Data Lake, and Blob Storage**.

TECHNICAL SKILLS

Languages: Python, Scala, R, SQL, SAS

Packages/Libraries: NumPy, Pandas, Matplotlib, Seaborn, ggplot2, SciPy, Scikit-learn, TensorFlow

Visualization Tools: Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP)

IDEs: Visual Studio Code, PyCharm, Jupyter Notebook

Database Management: MySQL, PostgreSQL, SQL, Oracle, Snowflake, dbt

Cloud Technologies: AWS, Azure

Methodologies: SDLC, Agile, Waterfall, Predictive Modeling

Version Control: Git, GitHub, Bitbucket

Data Manipulation: Data Analysis, Data Mining, Data Preprocessing, Data Mapping, Data Cleaning, Data Visualization, Data Modeling, Data Warehousing, Data Storytelling, Data Wrangling, Data Acquisition, Data Integration, Data Transformation

Other Skills: SSIS, SSRS, Machine Learning Algorithms, CI/CD, Probability Distributions, Confidence Intervals, ANOVA, Association Rules, Clustering, Advanced Analytics, ETL Processes, Informatica, Report Generation, Regression, A/B Testing, Forecasting & Modeling, Hypothesis Testing, Time Series Analysis, Root Cause Analysis

WORK EXPERIENCE

Renasant Bank, MS | Data Analyst

Nov 2024 - Present

- Delivered a **compliance analytics solution** for suspicious activity monitoring, supporting **alert triage** and **case prioritization** across high-volume banking transactions, with **audit-ready reporting** for compliance leadership.
- Established **Amazon S3** landing zones and orchestrated **AWS** batch processing for **2-4M transactions/day**, enabling lineage/backfills and publishing investigation datasets within **1-2 hours**, maintaining **88-94%** SLA adherence.
- Implemented **Apache Spark** to scale **distributed processing** for high-volume transaction transformations and feature prep, accelerating **investigation-ready datasets** and enabling faster **alert triage** and **case prioritization** refresh cycles.
- Built governed transformation layers in **dbt** to publish **alert, case, and cohort marts**, reducing KPI definition defects by **15-20%** by enforcing **tests** for nulls, uniqueness, and key integrity.
- Built **Alteryx Designer** workflows blending transaction, customer, and merchant data, cutting prep time **20-30%** while applying standardization, deduping, and validation rules to publish investigation-ready extracts.
- Developed Python **EDA** and feature **engineering scripts** to derive velocity, frequency, and time-window signals from transaction history, creating reusable risk attributes that strengthened alert prioritization and cohort-level analysis.
- Designed **Tableau** dashboards for alert volume, backlog, typologies, and cohort risk, improving review cycles **10-20%** via drill-down filters, exception views, and KPI trend tracking for compliance leaders.
- Implemented **Scikit-learn** alert prioritization models using **18-24 months** outcomes history, improving ranking by **25-35%** and reducing review effort by **20-30%**, with **SOX ITGC** traceable deployments.
- Performed **SAS data mining** to identify typology clusters and noise contributors, driving tuning recommendations that reduced low-value alert volume by **8-12%** and improved queue focus.
- Validated tuning changes using **ANOVA** and **hypothesis testing** across risk cohorts, confirming statistically significant gains and delivering **6-10%** precision improvement while keeping false positives controlled.

- Established an end-to-end analytics solution to support suspicious activity monitoring by consolidating transaction data and delivering KPI-driven dashboards to improve alert triage, investigation efficiency, and compliance outcomes.
- Automated extraction and validation of **500K+ daily transaction records** using basic **Pandas** checks, improving data accuracy, reducing manual reconciliation work, and accelerating analytics readiness for compliance analytics outputs.
- Produced scalable refresh workflows using **SSIS** to load and validate **20–30 reporting tables** per run, ensuring stable processing and reliable availability of curated datasets for monitoring dashboards.
- Deployed KPI reporting through **Power BI** dashboards to track alert volumes, high-risk merchants, regional trends, and case backlog trends, reducing turnaround time by **15%** for stakeholder review meetings.
- Executed structured data profiling and quality checks using **Advanced Excel** (Pivot Tables, VLOOKUP) and validation queries, identifying duplicates, null values, and incorrect formats across **50+ fields** affecting risk metrics.
- Created and maintained rule-based monitoring logic leveraging **Azure** data workflows to flag high-value spikes, rapid repeat transactions, and abnormal time-window behavior, improving alert tagging consistency for investigation review cycles.
- Constructed reusable reporting datasets through structured **data cleaning**, standardization, and transformation aligned with monitoring definitions, improving KPI accuracy and enabling repeatable trend tracking across business units.
- Performed **A/B testing** and trend analysis using summary tables and interactive visuals to evaluate rule effectiveness, identify high-risk segments, and support operational prioritization with clear follow-up actions.
- Managed version control of scripts and dashboard updates using **Git/GitHub**, improving traceability, reducing clarification cycles by **25%**, and supporting controlled promotion of reporting changes.
- Collaborated with compliance, QA, and business teams in an **Agile SDLC** environment, supporting sprint ceremonies, defect triage, and UAT validation to ensure timely releases and stable reporting enhancements.

PROJECT

Project 1: Vendor Performance Data Analytics Project

Technologies & Skills Used: SQL (Joins, CTEs, Aggregations), Python (Pandas, NumPy, EDA), Power BI (DAX, Interactive Dashboards), Data Cleaning, KPI Development, Data Visualization, Business Reporting, Stakeholder Communication.

- Analyzed vendor performance using **SQL** (Joins, CTEs, Aggregations) to extract and transform procurement data, evaluating supplier efficiency through KPIs such as on-time delivery rate, defect rate, and cost variance.
- Performed **data cleaning and exploratory data analysis** using **Python (Pandas, NumPy)** to identify delivery delays, pricing anomalies, and underperforming vendors, supporting root-cause insights and improvement recommendations.
- Built interactive **Power BI dashboards** using **DAX** to visualize vendor trends, scorecards, and KPI drilldowns, enabling clear business reporting and stakeholder-ready insights for procurement decision-making.

Project 2: Retail Customer Behavior & Shopping Trends Analytics Project

Technologies & Skills Used: Python (Pandas, NumPy, EDA), SQL (Joins, CTEs, Window Functions), Power BI (DAX, Data Modeling, Dashboards), Data Cleaning, Customer Analytics, KPI Development, Business Reporting, Data Visualization.

- Extracted and analyzed retail transaction and customer datasets using **SQL** (Joins, CTEs, Window Functions) to identify purchasing trends, repeat customer behavior, and revenue contribution across product categories and time periods.
- Performed **data cleaning and exploratory data analysis** using **Python (Pandas, NumPy)** to uncover shopping frequency patterns, basket trends, and high-value customer segments, supporting targeted marketing insights.
- Improved interactive Power BI dashboards using DAX and data modeling to track KPIs such as average order value, customer retention, and sales trends, delivering executive-ready reporting to support business decisions.

EDUCATION

Master of Science in Information Technology

- Concordia University Wisconsin, USA

Bachelor of Technology in Information Technology

- Vignana Bharathi institute of technology